

Plain-UDP Transport

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Revision: 1 Last modified: 2026-06-25T00:00:00Z

Master technical specification — Volume 2 (Data Plane), nano-detail document. Deepens the *Plain-UDP Transport* section (§3.2) of [01-data-plane §3.2]. This is a SPEC — it describes the `helix-transport::plain_udp` module to be built, not the shipping product. Source evidence cited inline by id: [01-DP §N] = `final/01-data-plane.md`; [04_ARCH §N] = `04_VPN_CLD/HelixVPN-Architecture-Refined.md`; [04_P0 §N] = `HelixVPN-Phase0-Spike.md`; [SYNTHESIS §N] = the cross-document synthesis; [10-QA §N] = `final/10-testing-acceptance-and-qa.md`. External WireGuard protocol constants (not part of this evidence base) are tagged [WG-PROTO] and, where their exact value is not load-bearing for this module, annotated honestly per constitution §11.4.6.

0. Scope & contract summary

`plain-udp` is the **default fast-path transport**: it carries already-encrypted WireGuard datagrams directly inside UDP, with **zero obfuscation**, over a single `tokio::net::UdpSocket` [01-DP §3.2, 04_P0 §4.3]. It is the lowest-latency, lowest-CPU carrier and the **throughput baseline** every other transport is measured against (the G1 Phase-0 gate: `through-tunnel iperf3` $\geq$ 80% of bare-link [04_P0 §G1, §8]).

This document owns, to nano-detail:

1. The `PlainUdp` type and its full Transport implementation (§3).
2. The on-the-wire byte layout — what `plain-udp` puts on L1, and why it treats the payload as opaque (I1) (§4).
3. The connection lifecycle state machine (§5).
4. **Failure detection that triggers ladder escalation** — the hard part, because UDP is connectionless and rarely errors on send (§6).
5. Error taxonomy, config knobs, edge cases, security, performance budget (§7–§11).
6. The concrete test points, tied to the §11.4.169 test-type taxonomy [10-QA §1] (§12).

Invariants inherited unchanged [01-DP §0.1]: I1 the transport never sees plaintext (it carries opaque WG datagrams); I2 the transport carries **unreliable datagrams**, never an ordered stream; I4 one transport crate, three consumers (client / connector / edge). plain-udp is the simplest possible witness to all three.

File: helix-transport/src/plain_udp.rs (the unit this doc specifies)

Trait: helix-transport/src/lib.rs::Transport (the contract; frozen Phase-0 interface)

Errors: helix-transport/src/error.rs::TransportError

1. When plain-udp is used — selection conditions

plain-udp is **ladder rung 0** in every TransportPolicy.order [01-DP §5.3]. The orchestrator selects it in these cases (closed set):

#	Condition	Source
C1	Unrestricted network, auto mode — order[0] == plain-udp; the default first attempt on every fresh connect.	[01-DP §5.3 "Default order"], [04_ARCH §3.2]
C2	Per-network memory hit — a prior successful plain-udp connect on this SSID / gateway fingerprint is remembered and re-tried first.	[01-DP §5.3 step 4], [04_P2 §1.4]
C3	Manual pin — user sets TransportPolicy.pin = PlainUdp, skipping the ladder entirely (Mullvad "WireGuard / automatic" UX).	[01-DP §5.3 step 1], [04_ARCH §3.2]
C4	Cooperative-leg policy — on the gateway↔connector leg of an asymmetric topology (D6), where the connector dials out of a cooperative network, plain-udp is the standing carrier.	[01-DP §11.3 D6], [SYNTHESIS §3 D6]
C5	Phase-0 spike — the only fully-implemented transport alongside masque-h3; the G1 vertical slice runs on plain-udp.	[04_P0 §4.3, §S0/§S1]

It is **escalated away from** (rung 0 → rung 1) only when its handshake-failure budget is exceeded (§6.4). It is never *de-escalated to* mid-session: a working tunnel that was established on a higher rung does not opportunistically downgrade to plain-udp without a fresh ladder walk (the per-network memory in C2 applies only at the next connect).

2. Module dependencies

Dependency	Role	Note
tokio (net, time, sync) bytes::Bytes	async UdpSocket, timers, Notify	the single async runtime [01-DP §3.2]

Dependency	Role	Note
	zero-copy datagram buffers	the Transport send/recv currency [01-DP §3.1]
async_trait	object-safe async Transport	matches the trait def [01-DP §3.1]
socket2	raw socket options (SO_SNDBUF, IP_MTU_DISCOVER, SO_BINDTODEVICE, recvmsg)	needed for the knobs in §11; tokio::net::UdpSocket::from_std adopts the configured fd
std::sync::atomic	lock-free HealthCell counters	hot-path health updates without a mutex (§3.3)

No obfuscation crate, no extra crypto crate — plain-udp adds **nothing** to the datagram. WG (helix-wg [01-DP §4]) is the only cryptography in the path.

3. Public API — full signatures

3.1 The PlainUdp type

```
// helix-transport/src/plain_udp.rs
use std::net::SocketAddr;
use std::sync::Arc;
use bytes::Bytes;
use tokio::net::UdpSocket;
use crate::{Transport, TransportError, TransportHealth};

/// The baseline carrier: one UDP socket, peer endpoint, opaque WG
/// datagrams (I1, I2).
/// Cheap to clone-share via `Arc`; the orchestrator holds it
/// behind `Box<dyn Transport>`.
pub struct PlainUdp {
    sock: Arc<UdpSocket>, // bound + (optionally) connected;
                        // see §3.4 / §11
    peer: SocketAddr, // gateway/connector endpoint (fixed
                    // unless §3.6 rebind)
    health: HealthCell, // §3.3 – lock-free liveness
                    // counters
    opts: UdpSocketOpts, // §11 – recorded so health/rebind
                    // logic can read them
}
```

Field rationale (nano-detail):

- sock: Arc<UdpSocket> — Arc (not bare UdpSocket) because send and recv run on **two independent task loops** (tun→wg→transport and transport→wg→tun) [01-DP §5.1]; both hold a clone. tokio::net::UdpSocket::send/recv take &self, so a single socket is shared

read+write concurrently with no lock — UDP send and recv on one fd are independent kernel operations.

- **peer**: `SocketAddr` — resolved from `TransportConfig::PlainUdp { peer, bind }` [01-DP §3.1], itself sourced from the `RouteMap's PeerRoute.endpoint_candidates[0]` pushed by the coordinator (doc 03) [01-DP §6.2]. v4 or v6 (§8 E5).
- **health**: `HealthCell` — the only state the *transport* surfaces upward; the WG handshake state lives in `helix-wg`, fused with this by the ladder (§6.3).

3.2 The Transport impl (complete)

This refines the Phase-0 sketch [01-DP §3.2, 04_P0 §4.3] into the production form: pooled recv buffer, RTT-unaware send, health marking, idempotent close.

```
#[async_trait::async_trait]
impl Transport for PlainUdp {
    async fn send(&self, dg: Bytes) -> Result<(), TransportError>
    {
        // Oversize guard FIRST (I-budget §4.4): never hand the
        // kernel a datagram the path
        // cannot carry – that yields a silent drop or EMSGSIZE
        // depending on DF (§8 E2).
        if dg.len() > self.opts.max_datagram {
            return Err(TransportError::Oversize(dg.len()));
        }
        let n = if self.opts.connect_peer {

            self.sock.send(&dg).await           // connected socket
            (ICMP errors surface, §6.2)
        } else {
            self.sock.send_to(&dg, self.peer).await
        };
        match n {
            Ok(_sent) => { self.health.mark_sent(); Ok(()) }
            Err(e)    => { self.health.mark_send_err();
                Err(map_io(e)) } // §7
        }
    }

    async fn recv(&self) -> Result<Bytes, TransportError> {
        // One MTU-sized scratch per call; pooled in the
        // production impl (§10 P4).
        let mut buf = vec![0u8; self.opts.recv_scratch]; //
        // default 1500 (§11)
        let n = if self.opts.connect_peer {
            self.sock.recv(&mut buf).await
        } else {
            // recv_from so we can drop datagrams whose source !=
            // peer (§8 E4, anti-spoof)
            let (n, from) = self.sock.recv_from(&mut
            buf).await.map_err(map_io)?;
            if !self.source_accepted(from) { return
            Err(TransportError::Closed /*see §6.2 note*/); }
            Ok::<usize, std::io::Error>(n)
        }
    }
}
```

```

        }.map_err(map_io)?;
        self.health.mark_recv(); // liveness clock reset (§6.2
        D2)
        buf.truncate(n);
        Ok(Bytes::from(buf))
    }

    fn kind(&self) -> &'static str { "plain-udp" }

    fn effective_mtu(&self) -> u16 { 1420 } // standard WG-over-
        IPv4 (§4.4) [04_P0 §4.3]

    fn health(&self) -> TransportHealth { self.health.snapshot() }

    async fn close(&self) -> Result<(), TransportError> {
        // UDP is connectionless: no FIN. Closing = dropping the
        // socket. Idempotent – a second
        // close is a no-op. The fd is released when the last Arc
        // clone drops (§5 Closed).
        Ok(())
    }
}

```

Honest correction vs the Phase-0 sketch. [01-DP §3.2] / [04_P0 §4.3] show `recv` returning on a bare `self.sock.recv(&mut buf)` with no source check and no oversize guard. That is a deliberately trivial spike body. The production `recv` here adds the §8 E4 source filter and the §3.2 `send` adds the §4.4 oversize guard — both required to avoid the bluff failure modes §6/§8 enumerate. The wire behaviour (an unmodified WG datagram in a UDP payload) is identical.

3.3 HealthCell — lock-free liveness

`TransportHealth` [01-DP §3.1] is `{ rtt_ewma_ms: Option<u32>, last_recv_age_ms: u64, send_errors: u32 }`. `plain-udp` populates it without a mutex on the hot path:

```

use std::sync::atomic::{AtomicU64, AtomicU32, Ordering};

pub(crate) struct HealthCell {
    last_recv_epoch_ms: AtomicU64, // monotonic ms since process
    // start of the last recv
    send_errors: AtomicU32,
    sent: AtomicU64, // counters (no per-packet log
    // – I5 no-logging)
    recvd: AtomicU64,
    rtt_ewma_ms: AtomicU32, // 0 == "unset" → mapped to
    // None in snapshot()
    start: std::time::Instant,
}

impl HealthCell {

```

```

fn mark_rcv(&self) {
    self.last_rcv_epoch_ms.store(self.now_ms(),
    Ordering::Relaxed);
    self.rcvd.fetch_add(1,
    Ordering::Relaxed); }
fn mark_sent(&self) { self.sent.fetch_add(1,
    Ordering::Relaxed); }
fn mark_send_err(&self) { self.send_errors.fetch_add(1,
    Ordering::Relaxed); }

/// RTT is NOT measured by plain-udp itself (it cannot – it
does not parse WG, I1).
/// `helix-wg` measures handshake RTT and pushes it here via
`set_rtt` (§6.3).
pub(crate) fn set_rtt(&self, ms: u32) {
    self.rtt_ewma_ms.store(ms.max(1), Ordering::Relaxed); }

fn snapshot(&self) -> TransportHealth {
    let r = self.rtt_ewma_ms.load(Ordering::Relaxed);
    TransportHealth {
        rtt_ewma_ms: if r == 0 { None } else { Some(r) },
        last_rcv_age_ms:
            self.now_ms().saturating_sub(self.last_rcv_epoch_ms.load(Ordering::Relaxed)),
        send_errors: self.send_errors.load(Ordering::Relaxed),
    }
}

fn now_ms(&self) -> u64 { self.start.elapsed().as_millis() as
    u64 }
}

```

Why RTT is set externally: plain-udp obeys I1 (never parses the payload), so it cannot tell a handshake-response datagram from a transport-data datagram. Liveness it *can* measure (last_rcv_age_ms = "a datagram of any kind arrived"). RTT it cannot — that requires matching a WG handshake-init to its response, which only helix-wg sees. helix-wg calls HealthCell::set_rtt on each completed handshake (§6.3). This keeps the I1 boundary clean.

3.4 Construction — the dial() entry & bind

dial(TransportConfig::PlainUdp { peer, bind }) [01-DP §3.1] resolves to:

```

impl PlainUdp {
    /// Bind, apply socket options, optionally connect(), return a
live carrier.
    /// Returns within the orchestrator's bounded dial timeout;
never blocks (§6.4).
    pub async fn dial(peer: SocketAddr, bind: SocketAddr, opts:
        UdpSocketOpts)
        -> Result<Self, TransportError>
    {
        // 1. Build the raw socket with socket2 so we can set DF /
buffers / SO_BINDTODEVICE
        // BEFORE the fd is adopted by tokio (some options must
precede bind).

```

```

    let domain = if bind.is_ipv6() { socket2::Domain::IPV6 }
    else { socket2::Domain::IPV4 };
    let raw = socket2::Socket::new(domain,
    socket2::Type::DGRAM, Some(socket2::Protocol::UDP))?;
    raw.set_nonblocking(true)?;
    apply_opts(&raw, &opts)?; // §11 –
    DF, buffers, bind-device, DSCP
    raw.bind(&bind.into())?; //
    EADDRINUSE → §7 / §8 E7
    if opts.connect_peer { raw.connect(&peer.into())?; } //
    surfaces ICMP errors (§6.2)
    let sock = UdpSocket::from_std(raw.into())?; // adopt
    into tokio
    Ok(Self { sock: Arc::new(sock), peer, health:
    HealthCell::new(), opts })
}
}

```

`dial()` itself does **no WireGuard handshake** — binding a UDP socket cannot fail for "endpoint blocked" the way a TCP/QUIC connect can. The *connect-time* failure surfaces later, at the WG layer, as a handshake that never completes (§6). This is the crucial asymmetry that shapes plain-udp's failure-detection design: **`dial()` succeeds almost always; the failure is discovered after handshake-init, not at dial.**

3.5 No TransportConfig secrets

`TransportConfig::PlainUdp { peer, bind }` carries **no SecretBytes** — unlike Shadowsocks (psk), Lw0 (session_key), or Hysteria2 (salamander_pw) [01-DP §3.1]. plain-udp has no transport-layer key material; all secrecy is WG's. This is a security property, not a gap (§9 S1).

3.6 Endpoint rebind (roaming)

The peer may change once Phase-2 NAT traversal fills multiple `endpoint_candidates` [01-DP §8]. plain-udp exposes a non-trait helper the orchestrator calls when WG roaming latches onto a new source address:

```

impl PlainUdp {
    /// Switch the peer endpoint without re-binding the local
    /// socket (keeps the source port
    /// stable so the gateway's NAT mapping survives). For
    /// connected sockets this re-issues
    /// connect(); for unconnected it just swaps `self.peer`.
    /// Cheap, lock-free under &mut.
    pub fn rebind_peer(&mut self, new_peer: SocketAddr) ->
    Result<(), TransportError> { /* ... */ }
}

```

Phase-0/Phase-1 never call this (single fixed gateway endpoint). It exists so the §8 E6 roaming edge case is handled by construction in Phase 2, not retrofitted.

4. Wire format — byte layout

plain-udp is the one transport whose wire bytes equal the WG bytes — there is **no framing of its own**. Understanding what is on the wire is what makes the security analysis (§9) and the failure-detection design (§6) precise.

4.1 L1/L2 stack (one user-bound WG datagram on plain-udp/IPv4)

```
+-----+
| IPv4 header    (20 bytes, no options)          |
← L1 (kernel)
+-----+
| UDP header     (8 bytes:  src port | dst port | len | csum)  |
← L1 (kernel)
+-----+
| WireGuard datagram (≤ 1452 bytes on a 1500-byte path)      |
← L2 payload = opaque to plain-udp (I1)
|  byte 0       : message type    (1)                      |
|  bytes 1..4   : reserved (zero) (3)                      |
|  [WG-PROTO] - informational only;                        |
|  bytes 4..    : type-specific body (handshake / transport) |
plain-udp NEVER reads these.
+-----+
```

- The **WG datagram is byte-identical** to what a stock WireGuard peer would emit. plain-udp copies it from Bytes into the UDP payload unmodified and back. That is the entire transform.
- Default endpoint: the gateway's WG UDP port (51820 in the Phase-0 rig; nft ... udp dport 51820 drop is the DPI block that triggers escalation [04_P0 §3, §S?]). The production peer.port() comes from the RouteMap, not hard-coded.

Packet flow through plain-udp (the two hot loops [01-DP §5.1]; the carrier is a thin opaque copy at the L2 boundary):

flowchart LR

```
    subgraph out["Outbound loop (tun → wg → transport)"]
        TUN1[TUN: plaintext IP pkt] --> WGE[helix-wg encrypt<br/>
>WgVerdict::WriteToTransport]
        WGE -->|opaque WG datagram, Bytes| GUARD{len ≤
max_datagram?<br/>§4.4}
        GUARD -- no --> OVR[Err Oversize<br/>orch lowers inner
MTU]
        GUARD -- yes --> SEND[PlainUdp::send<br/>UdpSocket send /
send_to]
        SEND --> WIRE1[(L1: IP+UDP+WG bytes<br/>to peer)]
    end
    subgraph inb["Inbound loop (transport → wg → tun)"]
        WIRE2[(L1: UDP datagram<br/>from network)] --> SRC{source
== peer?<br/>§8 E4}
        SRC -- no --> DROP[drop + counter<br/>keep looping]
        SRC -- yes --> RECV[PlainUdp::recv<br/>mark_recv →
carrier_live]
```



```

    RECV -->|opaque WG datagram| WGD[helix-wg decrypt<br/>
>WriteToTun / handshake]
    WGD --> TUN2[TUN: plaintext IP pkt]
end
classDef crypto fill:#eef,stroke:#447;
class WGE,WGD crypto;

```

The only plain-udp-owned steps are the **Oversize guard**, the **source filter**, and the **1:1 byte copy** to/from the socket — everything cryptographic (blue) is helix-wg's, honouring I1.

4.2 The payload is opaque (I1) — what plain-udp does and does not do

plain-udp DOES	plain-udp DOES NOT
copy the WG Bytes into one UDP datagram (1:1, no fragmentation, no coalescing)	parse, decrypt, validate, or reorder the WG payload
enforce a max size (Oversize guard, §4.4)	add length prefixes, headers, padding, or obfuscation
drop datagrams from a non-peer source (§8 E4)	maintain per-flow or per-packet durable state (I5)
reset the liveness clock on any recv	distinguish handshake vs transport-data (that is WG's job)

4.3 WG message taxonomy (informational — transport does NOT branch on it)

These are **external WireGuard protocol constants** [WG-PROTO], included only so the §6 failure analysis is concrete. plain-udp treats all four identically (opaque payload, I1). Exact byte sizes are protocol facts outside this evidence base; they are **not load-bearing for this module** and any consumer that needs them parses WG inside helix-wg, never here:

type (byte 0)	Meaning	Approx. fixed size	Relevance to plain-udp
1	Handshake Initiation	UNVERIFIED: ~148 B [WG-PROTO]	the <i>first</i> datagram plain-udp sends after dial; its loss-without-response is the core failure signal (§6)
2	Handshake Response	UNVERIFIED: ~92 B [WG-PROTO]	the <i>first</i> inbound datagram on a healthy connect; its arrival = liveness + RTT sample (§6.3)
3	Cookie Reply	UNVERIFIED: ~64 B [WG-PROTO]	load/DoS defence; opaque to plain-udp
4	Transport Data	variable, 32 B WG overhead (4 type+reserved)	steady-state payload; the 32-B figure is the grounded basis of the

type (byte 0)	Meaning	Approx. fixed size	Relevance to plain-udp
		$\cdot 4 \text{ receiver-index} \cdot 8$ counter $\cdot 16$ Poly1305 tag)	MTU budget §4.4 [04_P0 §4.3 via 01-DP §10]

The 32-byte transport-data overhead is grounded — it is the 32 WG hdr/tag term in the $1500 - 20 \text{ IP} - 8 \text{ UDP} - 32 = 1420$ MTU derivation [04_P0 §4.3, 01-DP §10]. The init/resp/cookie sizes are marked UNVERIFIED per §11.4.6 because they are not in the evidence base and are not needed by this module.

4.4 MTU & overhead budget

plain-udp adds **zero** transport overhead on top of WG, so it reports the highest `effective_mtu()` of any carrier [01-DP §10]:

`effective_mtu = 1420` (IPv4: $1500 - 20 \text{ IP} - 8 \text{ UDP} - 32 \text{ WG hdr/tag} \approx 1420$) [04_P0 §4.3]

Path	L1 MTU	IP hdr	UDP hdr	WG hdr/tag	inner WG MTU
IPv4 (typical)	1500	20	8	32	1420 [04_P0 §4.3]
IPv6	1500	40	8	32	UNVERIFIED: ~ 1400 (derived: $1500 - 40 - 8 - 32$; not asserted in evidence base)
IPv6 min-MTU floor	1280	40	8	32	UNVERIFIED: ~ 1200 (derived)

Budget rules [01-DP §10]:

1. **Inner WG MTU = `min(effective_mtu(), path-MTU-discovered)`** — the orchestrator sets WG's MTU from `effective_mtu()` clamped by PMTU (§11 `set_df = true` enables honest PMTUD).
2. `max_datagram` (the `0versize` threshold, §3.2) = `effective_mtu() + 32` (the WG header/tag), i.e. the largest WG datagram a 1420-byte inner MTU can produce $\approx$ **1452 bytes** on IPv4. A larger datagram is a hard `TransportError::0versize` — never silently truncated (§8 E2).
3. plain-udp performs **no L3 fragmentation**; if the path MTU drops below the configured inner MTU, the orchestrator lowers the inner WG MTU (it does not ask the transport to fragment).

5. Connection lifecycle state machine

plain-udp's own lifecycle is intentionally tiny — it has no handshake of its own. The *tunnel* state (Connecting/Handshaking/Connected/...) is the orchestrator's TunnelStatus [01-DP §5.2]; plain-udp contributes the **carrier** sub-states below, observed via health().

```
stateDiagram-v2
    [*] --> Unbound
    Unbound --> Bound: dial() – bind + apply opts + (connect)
    Bound --> Live: first send() succeeds (WG handshake-init out)
    Live --> Healthy: recv() observed (last_recv_age_ms small) – a
    datagram came back
    Healthy --> Live: last_recv_age_ms grows past keepalive
    interval (no recv yet, still ok)
    Live --> Stale: last_recv_age_ms > liveness_timeout AND
    handshake not complete
    Healthy --> Stale: last_recv_age_ms > liveness_timeout (peer
    went silent mid-session)
    Stale --> Live: a recv() arrives again (transient loss
    recovered)
    Bound --> Errored: send() returns EHOSTUNREACH/ECONNREFUSED
    (connected socket, ICMP)
    Live --> Errored: send() returns ICMP-derived error (connected
    socket)
    Errored --> [*]: close() – drop socket; ladder escalates
    Stale --> [*]: failure budget exceeded → close() → ladder
    escalates (§6.4)
    Healthy --> Closed: close() (orchestrator teardown / pin
    change)
    Live --> Closed: close()
    Closed --> [*]
```

State meanings (what the orchestrator reads from TransportHealth):

Carrier state	Detected by	Tunnel-level meaning
Bound	dial() returned 0k	socket exists; nothing sent yet
Live	sent > 0, recvd == 0, send_errors == 0	handshake-init(s) out, awaiting response — the critical window (§6)
Healthy	recvd > 0, last_recv_age_ms < keepalive	datagrams flowing both ways
Stale	last_recv_age_ms > liveness_timeout	peer silent; transient or terminal (§6.2 D2)
Errored	send_errors > 0 from an ICMP-derived io::Error (connected socket)	endpoint actively rejecting (§6.2 D3)
Closed	close() / Arc dropped	carrier gone

6. Failure detection that triggers escalation

This is the heart of the document. The escalation ladder [01-DP §5.3] walks to the next rung on `DialTimeout` / `HandshakeFailed` / `EndpointBlocked` exceeding a per-rung failure budget. For plain-udp specifically, **`dial()` almost never fails** (§3.4) — so the failure signal must be synthesised from *liveness* + *WG handshake progress*, not from a send error.

6.1 Why UDP send rarely errors (the connectionless trap)

A bare (unconnected) `UdpSocket::send_to` returns `Ok` as soon as the datagram is queued to the kernel — even if the destination is a black hole. There is no SYN/ACK, no connection refusal at the transport. Consequently:

- A DPI block that silently drops udp `dport 51820` [04_P0 §3] produces **zero send errors**. plain-udp's send keeps returning `Ok`; the only symptom is that **no response ever arrives**.
- Therefore plain-udp's failure detector is fundamentally a **timeout-on-no-recv** detector, optionally sharpened by **ICMP errors on a connected socket**.

6.2 The detection signals (closed set)

ID	Signal	Source	Maps to
D1	Dial failure — bind/connect syscall error (rare: <code>EADDRINUSE</code> , bad bind addr)	<code>dial()</code> (§3.4)	<code>TransportError::Io</code> → orchestrator may retry bind or <code>DialTimeout</code>
D2	Liveness timeout — <code>health().last_recv_age_ms > liveness_timeout_ms</code> while the WG handshake has not completed	<code>HealthCell</code> + <code>helix-wg</code> handshake state (§6.3)	synthesised <code>HandshakeFailed("no response")</code>
D3	ICMP rejection — on a connected socket (<code>opts.connect_peer = true</code>), an ICMP <i>port/host/net unreachable</i> surfaces as send/recv returning <code>ECONNREFUSED</code> / <code>EHOSTUNREACH</code> / <code>ENETUNREACH</code>	kernel → <code>io::Error</code> (§3.2)	<code>EndpointBlocked</code> (D3 is a <i>fast</i> signal — no need to wait the full liveness timeout)
D4	Source mismatch flood — many inbound datagrams whose source $\neq$ peer (spoof / wrong-NAT)	<code>recv_from</code> filter (§8 E4)	counter only; does not by itself escalate, but suppresses false "Healthy"
D5	Persistent send error — <code>EMSGSIZE</code> (DF + oversize), <code>ENOBUFS</code> , <code>EPERM</code> (firewall)	<code>send</code> <code>io::Error</code> (§3.2)	<code>Io</code> ; repeated → budget toward escalation

D3 is why the spec recommends `connect_peer = true` for the fixed-gateway client leg (§11): it converts a silent black hole into a *fast, explicit* `EndpointBlocked`, cutting escalation latency from `liveness_timeout` (seconds)

to one RTT. The trade-off: a connected socket pins the peer, so it must be `rebind_peer'd` (§3.6) on Phase-2 roaming. For the connector leg (cooperative network, fixed gateway) `connect_peer = true` is strictly better.

6.3 Fusing carrier liveness with WG handshake state (the I1-clean seam)

Because plain-udp cannot see WG (I1), the *authoritative* "is the tunnel up?" verdict is the WG handshake state in helix-wg, not the carrier. The orchestrator fuses them:

```
tunnel_up == helix_wg.handshake_complete()           // WG says
a session key exists
carrier_live == health().last_rcv_age_ms < liveness_timeout //
plain-udp says bytes flow
```

- helix-wg emits `WgVerdict::WriteToTransport(handshake_init)` on connect; the orchestrator sends it via plain-udp. plain-udp marks sent.
- If a response arrives, `recv` yields it, plain-udp marks `recv` ($\rightarrow$ `carrier_live`), and `helix-wg.handle_transport_in` completes the handshake ($\rightarrow$ `tunnel_up`) and reports RTT, which the orchestrator feeds to `HealthCell::set_rtt` (§3.3).
- If **no** response arrives within `liveness_timeout` AND ! `helix_wg.handshake_complete()`, that is **D2** $\rightarrow$ the orchestrator synthesises `HandshakeFailed("plain-udp: no response")` and charges the rung's failure budget.

This division is deliberate: plain-udp stays a dumb opaque carrier; the smart "did the crypto handshake succeed?" decision lives where the crypto lives.

6.4 The failure budget & escalation trigger

The rung's `FailureBudget { max_handshakes: u8, window: Duration }` [01-DP §5.3] is evaluated by the ladder, fed by plain-udp's signals:

```
on each connect attempt for rung=plain-udp:
  send WG handshake-init (retry per WG's own backoff, ~5 s ×
max_handshakes)
  await EITHER:
    tunnel_up                                $\rightarrow$  Connected{transport:"plain-
udp", rtt}      (emit, remember §C2)
    D3 (ICMP EndpointBlocked)                $\rightarrow$  immediate budget hit (fast
path)
    D2 (liveness_timeout, no resp)           $\rightarrow$  budget hit
  if budget exceeded (handshakes  $\geq$  max_handshakes within window):
    close() this PlainUdp; orchestrator emits Reconnecting; ladder
 $\rightarrow$  rung 1 (lwo) [01-DP §5.3]
```

Recommended Phase-1 defaults (project-tunable knob, §11): `max_handshakes = 3`, `window = 15 s`, `liveness_timeout_ms = 4000`. These mirror "a few seconds before escalating" [04_ARCH §3.2 "after N failed handshakes"] and are calibrated, not hard-coded, against the Phase-0 G1 rig (§12) per §11.4.6/§11.4.107(13).

6.5 Sequence — plain-udp failure → escalation to rung 1

```
sequenceDiagram
    autonumber
    participant Orch as Orchestrator (helix-core)
    participant Wg as helix-wg (Tunn)
    participant Tx as PlainUdp (rung 0)
    participant Net as Network / DPI
    participant Lad as Ladder (TransportPolicy)

    Orch->>Tx: dial(PlainUdp{peer=gw:51820, connect_peer=true})
    Tx-->>Orch: Ok (bound + connected) %% dial almost never fails (§3.4)
    Orch->>Wg: start handshake
    Wg-->>Orch: WriteToTransport(handshake_init #1)
    Orch->>Tx: send(init#1)
    Tx->>Net: UDP :51820 (opaque WG bytes)
    Net--xTx: DROP (nft udp dport 51820 drop) %% silent black hole

    alt connected socket → ICMP arrives (D3, fast)
        Net-->>Tx: ICMP port-unreachable
        Tx-->>Orch: send()/recv() = Err(EndpointBlocked)
        Orch->>Lad: budget hit (fast)
    else no ICMP → liveness timeout (D2)
        Note over Tx,Orch: no recv; health.last_recv_age_ms climbs
        Wg-->>Orch: WriteToTransport(handshake_init #2, #3) %% WG retries ~5s apart
        Orch->>Tx: send(init#2), send(init#3)
        Tx->>Net: dropped each time
        Orch->>Orch: liveness_timeout 4s, !handshake_complete → synth HandshakeFailed (D2)
        Orch->>Lad: budget hit x3 within 15s window
    end

    Lad->>Lad: max_handshakes exceeded → escalate rung 0 → rung 1
    Orch->>Tx: close() %% drop socket
    Orch-->>Orch: emit TunnelStatus::Reconnecting
    Lad->>Orch: next = lwo (rung 1) [01-DP §5.3]
    Note over Orch,Net: ladder continues lwo → masque-h3 → shadowsocks → udp-over-tcp
```

7. Error taxonomy & mapping

plain-udp produces a strict subset of `TransportError` [01-DP §3.1 error.rs]. The `map_io` helper classifies `std::io::Error`:

```
fn map_io(e: std::io::Error) -> TransportError {
    use std::io::ErrorKind::*;
    match e.kind() {
```

```

    ConnectionRefused                =>
    TransportError::EndpointBlocked, // ICMP port-unreach (D3)
    HostUnreachable | NetworkUnreachable =>
    TransportError::EndpointBlocked, // ICMP net/host (D3)
    // EMSGSIZE has no stable ErrorKind on all platforms →
    // matched via raw_os_error below
    _ if is_emsgsize(&e)                =>
    TransportError::Oversize(0),         // DF + too big (D5/E2)
    _                                    =>
    TransportError::Io(e),               // ENOBUFS, EPERM, ...
  }
}

```

TransportError	Cause on plain-udp	Triggers escalation?	Detect signal
DialTimeout	dial() exceeded the orchestrator's bound dial timeout (rare; bind hang)	yes	D1
HandshakeFailed(String)	synthesised by orchestrator on D2 (liveness timeout, no WG response)	yes (budget)	D2
EndpointBlocked	ICMP unreachable on connected socket	yes (fast)	D3
Oversize(usize)	datagram > max_datagram; or EMSGSIZE with DF set	no — caller bug / PMTU; orchestrator lowers inner MTU (§4.4)	D5/E2
Closed	recv after close(); or source-mismatch sentinel (§3.2 note)	n/a (teardown)	—
Io(io::Error)	ENOBUFS, EPERM (firewall), EADDRINUSE (bind)	repeated → budget	D1/D5
Quic(_)	never — plain-udp has no QUIC	—	—

Note (§3.2 source-mismatch): returning Closed for a spoofed-source datagram is a placeholder; the production impl **loops internally** (drops the bad datagram, keeps awaiting a good one) rather than surfacing an error, so one spoofed packet cannot tear the tunnel. Treat the inline return Err(Closed) in §3.2 as shorthand for "continue the recv loop" — captured as test point T-EDGE-4 (§12).

8. Edge cases (closed enumeration)

ID	Edge case	Required behaviour	Test
E1	0-length datagram received	accept; <code>Bytes::new()</code> ; mark <code>recv</code> (a WG keepalive can be tiny but is ≥ 32 B — a true 0-length is a probe, still counts as liveness)	T-EDGE-1
E2	Overflow on send (<code>dg.len() > max_datagram</code>)	hard <code>Overflow</code> BEFORE <code>syscall</code> ; never truncate; never split	T-EDGE-2 (UNIT)
E3	recv buffer too small (<code>n == recv_scratch</code> , possible truncation)	the WG payload $\leq$ inner MTU < <code>recv_scratch=1500</code> , so truncation cannot occur for in-spec datagrams; an <code>n == scratch</code> event is logged as a counter anomaly (over-MTU peer) and the datagram dropped (WG would reject a truncated tag anyway)	T-EDGE-3
E4	Source-address mismatch (<code>recv_from</code> source $\neq$ peer)	drop + counter (D4); do not mark <code>recv</code> (else a spoofer keeps <code>carrier_live</code> falsely true); keep looping	T-EDGE-4 (SEC)
E5	Dual-stack v4/v6 peer	bind domain follows <code>bind.is_ipv6()</code> ; a v6 bind with <code>IPV6_V6ONLY=false</code> can reach v4-mapped peers (§11 knob); <code>effective_mtu</code> adjusts (§4.4)	T-EDGE-5
E6	Peer roams (Phase 2, new candidate)	<code>rebind_peer</code> (§3.6); keep local port stable so gateway NAT mapping survives	T-EDGE-6 (Phase 2)
E7	EADDRINUSE on bind	surface Io; orchestrator picks an ephemeral port (<code>bind 0.0.0.0:0</code>) and retries once before <code>DialTimeout</code>	T-EDGE-7
E8	ENOBUS under send burst	Io; the send loop applies bounded backpressure (<code>await writable()</code>), never busy-spins; counts toward D5 only if persistent	T-EDGE-8 (STRESS)
E9	Socket closed under an in-flight recv	<code>recv</code> returns <code>Closed</code> ; the <code>recv</code> loop exits cleanly (cancel-safe — <code>recv</code> is <code>select!-usable</code> per the trait contract [01-DP §3.1])	T-EDGE-9 (RACE)
E10	DF set, path MTU shrinks	<code>EMSGSIZE</code> $\rightarrow$ <code>Overflow</code> ; orchestrator lowers inner WG MTU and the next send fits (§4.4 rule 3)	T-EDGE-10

9. Security considerations

plain-udp is the **least private** transport by design — it is the fast path for *unrestricted* networks, not the evasion path. Its security posture:

ID	Property	Detail
S1	No transport secrets	<code>TransportConfig::PlainUdp</code> carries no key (§3.5). Nothing to leak at the transport layer; WG owns all confidentiality/integrity (Noise IK, ChaCha20-Poly1305) [01-DP §4, SYNTHESIS §2].
S2	Trivially DPI-fingerprintable	the wire bytes ARE WireGuard (byte 0 = WG message type, fixed handshake sizes) [WG-PROTO]; a censor's <code>udp dport 51820 drop</code> or a WG-signature rule blocks it instantly [04_P0 §3]. This is <i>expected</i> — when blocked, the ladder escalates to an obfuscating rung (<code>lwo/masque-h3</code>) [01-DP §5.3]. <code>plain-udp</code> must not be presented to the user as censorship-resistant.
S3	Source-spoof resistance	the <code>recv_from</code> source filter (E4) drops datagrams not from peer; combined with WG's cryptographic peer authentication, a spoofed source cannot inject into the tunnel — it can at most waste a buffer. No amplification: <code>plain-udp</code> never replies to an unsolicited datagram (it only echoes WG, which itself does not amplify).
S4	Kill-switch interface pin	<code>opts.bind_device = Some("wg-kill0")</code> (<code>S0_BINDTODEVICE</code>) binds the socket to the protected interface so a route flap cannot leak cleartext onto the physical NIC. The kill-switch state machine (security doc) drives this; <code>plain-udp</code> only honours the option.
S5	No-logging by construction (I5)	<code>HealthCell</code> holds only aggregate atomics (<code>sent/recvd/send_errors</code>) — never a per-packet or per-connection durable record [01-DP §0.1 I5, SYNTHESIS §7].
S6	PMTU/DF honesty	<code>set_df = true</code> (§11) prevents on-path fragmentation that could be a side channel and yields honest <code>EMSGSIZE</code> (E10) rather than silent black-holing.
S7	ICMP trust	D3 trusts ICMP unreachable to escalate <i>faster</i> , but ICMP is unauthenticated. A forged ICMP could force a premature escalation (DoS-to-slow-path, not a confidentiality break). Mitigation: ICMP only accelerates an escalation that the liveness timeout (D2) would reach anyway; it never causes a <i>downgrade</i> or key exposure. Recorded as an accepted residual risk (§11.4.6 — stated, not hidden).

10. Performance budget

`plain-udp` is the **baseline** — every other transport is measured as a fraction of it [01-DP §3.2, 04_P0 §8].

ID	Metric	Target	Source
P1	through-tunnel throughput (TCP+UDP iperf3)	≥ 80% of bare-link (the G1 gate)	[04_P0 §G1, §8]
P2	added latency vs bare path (ping overlay)	< 2 ms added, p50 and p99	[04_P0 §8 table]
P3	CPU	lowest of all transports (no obfuscation, no extra crypto)	[01-DP §3.2, 04_ARCH §3.2]
P4	per-recv allocation	one pooled MTU buffer; production uses a BytesMut slab + recvmmsg batch (§11 recv_mmsg_batch) to amortise syscalls — the §3.2 vec! [0u8; 1500] per call is the spike form, replaced by a pool	derived [04_P0 §8 "CPU/ Gbps"]
P5	handshake-to- Connected	one RTT after a non-blocked path (no transport handshake to add)	derived (§6)

Performance levers (nano-detail, all §11 knobs):

- **SO_SNDBUF / SO_RCVBUF** sized to $\geq$ BDP so bursts do not ENOBUFS (E8) / drop on recv.
- **GSO/GRO** (Linux UDP_SEGMENT / UDP_GRO): batch multiple WG datagrams per syscall on the edge's high-fan-in path. UNVERIFIED whether Phase-0/1 enables it — flagged as a Phase-2 edge-throughput lever, not asserted as shipped.
- **recvmmsg** (P4) batches inbound datagrams; the recv loop yields a Vec<Bytes> internally and feeds them to WG one at a time (preserving I2 unreliable-datagram semantics).

11. Configuration knobs — UdpSocketOpts

```
#[derive(Clone, Debug)]
pub struct UdpSocketOpts {
    pub send_buf_bytes:    usize,           // SO_SNDBUF      –
                          default 2 MiB
    pub recv_buf_bytes:    usize,           // SO_RCVBUF      –
                          default 4 MiB (edge: larger)
    pub recv_scratch:      usize,           // per-recv buffer –
                          default 1500 (≥ max_datagram)
    pub max_datagram:      usize,           // Oversize guard  –
                          default effective_mtu()+32 = 1452
    pub set_df:            bool,            // IP_MTU_DISCOVER=D0 /
                          IP_DONTFRAG – default true (PMTUD)
    pub connect_peer:      bool,            // UdpSocket::connect()
                          → ICMP errors (D3) – default true (fixed gw)
    pub bind_device:       Option<String>, // SO_BINDTODEVICE –
                          kill-switch pin (S4) – default None
    pub dscp:              Option<u8>,     // IP_TOS / IPV6_TCLASS
                          – default None
}
```

```

pub v6_only:
    bool,          // IPV6_V6ONLY – default false (dual-
                    stack, E5)
pub recv_mmsg_batch: usize,          // recvmsg batch
    (Linux) – default 64
// --- ladder-facing (read by the orchestrator, not the
    socket) ---
pub liveness_timeout_ms: u64,        // D2 – default 4000
pub max_handshakes:      u8,        // failure budget –
    default 3 [04_ARCH §3.2]
pub budget_window_ms:
    u64,          // failure budget window – default 15000
}

```

Every numeric default is **calibrated against the Phase-0 G1 rig**, not adopted from literature, per §11.4.6 / §11.4.107(13). They are project knobs (§11.4.35) the coordinator may override per-network via the pushed TransportPolicy [01-DP §5.3].

12. Test points — tied to §11.4.169 [10-QA §1]

Each row is a workable-item test obligation; abbreviations are the [10-QA §1] taxonomy. Every PASS cites captured evidence (§11.4.5/.69/.107), never config-only.

ID	Test point	§11.4.169 type	Evidence / harness
T-UNIT-1	effective_mtu() == 1420; max_datagram == 1452	UNIT	cargo test assert
T-UNIT-2	Oversize guard: send(1453 B) → Err(Oversize(1453)), no syscall	UNIT	cargo test (mock socket)
T-UNIT-3	map_io classification table (ECONNREFUSED→EndpointBlocked, EMSGSIZE→Oversize, ...)	UNIT	table-driven cargo test
T-UNIT-4	HealthCell snapshot math: last_recv_age_ms, rtt None?Some, atomic counters	UNIT	cargo test
T-S0	loopback echo: dial() two PlainUdp, send/recv round-trips a WG-sized blob (the S0 milestone)	INT	netns loopback [04_P0 §S0]
T-G1	through-tunnel iperf3 ≥ 80% bare-link + ping LAN host succeeds (the G1 gate, throughput baseline)	BENCH/ E2E/PERF	netns+netem rig, bench.sh [04_P0 §G1, §8]
T-FAIL-D2	DPI drop (nft udp dport 51820 drop) → no response → D2 liveness timeout → synth HandshakeFailed → ladder escalates rung 0→1; captured as an ordered TunnelStatus trace	E2E/SEC	netns + nftables [04_P0 §3]
T-FAIL-D3		E2E	netns reject rule

ID	Test point	§11.4.169 type	Evidence / harness
T-EDGE-1..10	connected socket + ICMP port-unreachable → EndpointBlocked within one RTT (faster than D2) the §8 edge cases (0-len, oversize, source-mismatch drop, dual-stack, EADDRINUSE, ENOBUFS, cancel-safe recv, DF/PMTU)	UNIT/ SEC/ RACE/ STRESS	per-row harness in §8
T-STRESS-1	≥ 100 iters / ≥ 30 s sustained + ≥ 10 parallel sockets; no leak, no ENOBUFS storm (§11.4.85)	STRESS	stress_chaos.sh ab_stress_* [10-QA §5.9]
T-CHAOS-1	kill the edge mid-iperf3; assert the tunnel recovers (reconnect/escalate) and no cleartext leaks during the gap	CHAOS	ab_chaos_kill_pid_during [10-QA §5.9]
T-RACE-1	concurrent send+recv on the shared Arc<UdpSocket>; cargo +nightly loom / tsan — no data race, no deadlock	RACE	loom model [10-QA §1]
T-SEC-1	source-spoof injection (E4): spoofed-source datagrams dropped, carrier_live not falsely set, WG never accepts	SEC	pcap + netns [10-QA §1]
T-REC-1	window-scoped MP4 of the G1 slice + media-validation verdict (per user-visible-claim, §11.4.154/.163)	REC	panoptic→vision_engine [10-QA §1]
T-CHAL-1	a challenges/helix_qa bank entry scoring PASS only on the captured G1 evidence (§11.4.27/.107)	CHAL/HQA	challenges submodule [10-QA §1]

DDOS and SCALE are NOT_APPLICABLE for plain-udp at MVP (single-node self-host topology, §11.4.3 topology-SKIP) and re-arm in Phase 2 [10-QA §1 table]. The plain-udp paired §1.1 meta-test mutation: invert the Oversize guard (> → <) and assert T-UNIT-2 FAILs — proving the guard test is not a bluff.

13. Frozen contract & out-of-scope

- **Frozen Phase-0 interface** [01-DP §3.1, 04_P0 §14]: the Transport trait method set (send/recv/kind/effective_mtu/health/close) is a frozen contract; PlainUdp's *implementation* (pooling, recvmmsg, GSO) may evolve, its *signatures* may not.
- **Out of scope here** (owned elsewhere): the Transport trait & dial() registry, the escalation-ladder algorithm, and TransportPolicy live in [01-DP §3.1 / §5.3]; WG handshake state and WgVerdict in helix-wg [01-DP §4]; the RouteMap that sources peer in helix-route [01-DP §6.2]; the obfuscating transports (lwo, masque-h3, shadowsocks, udp-over-tcp) in their own v02 docs. The kill-switch state machine (which drives bind_device, S4) is referenced for its seam only and specified in the security doc.

End of transport-plain-udp.md (Volume 2 — Data Plane). Pair with 01-data-plane.md §3.2 (the parent section this deepens), the transport-masque-h3.md nano-spec (the rung-2 obfuscating counterpart whose MTU/CPU penalty is measured against this baseline), and 10-testing-acceptance-and-qa.md §1 (the §11.4.169 test-type contract the §12 points instantiate).